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• github.com/MatheusPBC

Leadership Activities

SmartLy

Brazil

Apoio tecnico, revisao arquitetural e mentoria pratica

Apr 2024 - Present

- Apoio em decisoes de arquitetura backend, revisao de solucoes e evolucao de fluxos serverless em producao.
- Mentoria pratica para analise de impacto, depuracao, modelagem de dados e manutencao de sistemas distribuidos.
- Construcão de padroes de investigacao com IA aplicada, RAG, MCP e agentes para acelerar entendimento de codebases complexas.

Experience

SmartLy

Brazil

Backend Developer and Architect

Apr 2024 - Present

- Architected and developed a cloud-native IoT device management platform using AWS AppSync, AWS Lambda, DynamoDB, GraphQL, and serverless integrations.
- Modeled high-volume telemetry data and flows, balancing millions of messages, latency, per-operation cost, fan-out, and operational consistency.
- Designed real-time communication alternatives using GraphQL subscriptions, MQTT, WebSockets, DynamoDB Streams, Redis, interest management, and selective fan-out.
- Refactored backend structures, queries, and data contracts to reduce coupling, improve maintainability, and unlock production system evolution.
- Built and maintained APIs, Lambdas, and event-driven integrations focused on observability, rollback, security, and predictable cost.
- Investigated bottlenecks, reviewed architecture, supported technical decisions, and mentored the team through practical engineering trade-offs.

Eletrisa

Brazil

Full Stack Developer

Aug 2023 - Apr 2024

- Developed APIs, web applications, and internal systems for customers in the energy and power plant sector.
- Built complete systems with Node.js, TypeScript, React, MongoDB, Docker, and automated integrations.
- Created APIs with full CRUD flows, validation, authentication, and structured operational data.
- Developed and maintained Node-RED flows, including dashboards, integrations, automation, and real-time data processing.
- Created pipelines to read field equipment data, persist raw data, validate, transform, and periodically send data to external integrations.
- Worked with telemetry and hydrology data such as rainfall, level, flow, timestamps, extra fields, and equipment/item grouping.
- Fixed MongoDB aggregation, query performance, timestamp handling, table rendering, and dashboard issues.

Intelligence 4 Innovation

Brazil

Systems Developer / DevOps

Feb 2022 - Jun 2023

- Developed systems, flows, and applications with strong use of databases and automation.
- Supported and evolved systems connected to a large database used in a public and municipal environment.
- Developed and adapted code for continuous maintenance, support, and improvement of existing systems.

- Worked with Python, C++, GeneXus, and SQL, deepening data modeling, queries, manipulation, and production problem solving.
- Supported DevOps routines, application maintenance, and technical process improvements.

Plasnox Ind. e Com. de Plásticos Ltda	Brazil
<i>Purchasing Assistant</i>	<i>Jan 2021 - Nov 2021</i>
• Issued invoices, registered purchases, and operated enterprise management systems. • Gained early experience with administrative processes, operational organization, and corporate routines.	

Skills

- **Languages:** Portuguese (Native), English (Advanced / Fluent)
- **Technical:** AWS serverless architecture, AWS AppSync, AWS Lambda, DynamoDB, DynamoDB Streams, GraphQL subscriptions, Event-driven architecture, IoT telemetry, High-volume data pipelines, Cost-aware cloud design, Observability, Backend architecture, Python, TypeScript, JavaScript, Node.js, Java, PHP, C#, Ruby, C++, SQL, Express, Flask, Django, CherryPy, REST APIs, GraphQL, JWT authentication, SOAP integrations, Serverless architecture, AWS Lambda, AWS AppSync, DynamoDB, DynamoDB Streams, Docker, Terraform, Pulumi, Git, CI/CD, MongoDB, MySQL, SQL Server, React, Next.js, Vue, Quasar, Pinia, Node-RED, RAG, MCP, LLMs, AI agents, Linux, Neovim, tmux

Education

Instituto Federal de Santa Catarina - IFSC	Santa Catarina, Brazil
<i>Análise e Desenvolvimento de Sistemas</i>	<i>Dec 2020 - Present</i>
• Formação superior em desenvolvimento de sistemas, arquitetura de software, bancos de dados e engenharia de software.	

Instituto Federal Catarinense - IFC	Santa Catarina, Brazil
<i>Técnico em Informática</i>	<i>Dec 2019 - Dec 2020</i>
• Formação técnica em informática concluída em 2020.	